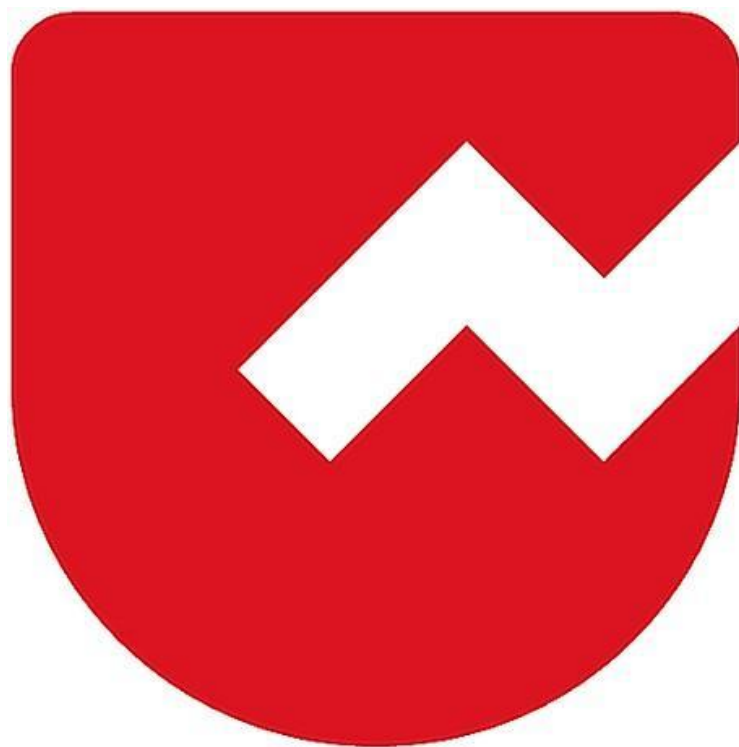




Zoho Schools for Graduate Studies



Notes

Constructor

Definition :

- A constructor is a special method inside a class.
- It is invoked implicitly at the time of object creation.
- It is used to initialize the state of an object.
- We can't invoke constructor like a normal method whenever required.

State of an Object :

- The value of the instance variables at a given instant of time is referred to as the state of an object.

Characteristics of Constructor :

- Cannot be invoked explicitly like a normal method.
- Does not have a return type.
- Shares the class name.

Purpose of Constructor :

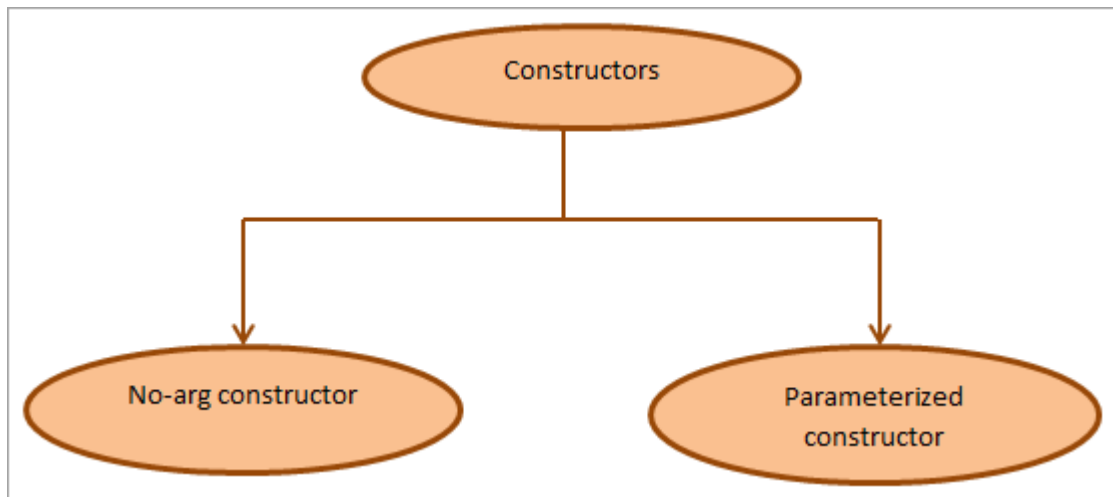
- Constructors are generally used to initialize state of an object

What is a Constructor?

- A special type of method invoked implicitly at the time of instantiation of an object, for the purpose of initializing the object.

Types of Constructors :

1. Parameterized Constructor → accepts arguments.
2. Non-Parameterized Constructor → no arguments.



Parameterized Constructor :

- A constructor that accepts arguments to initialize the object.

Non - Parameterized (or) Parameterless Constructor:

- A constructor that does not accept arguments.
- Supplied by default only if there is no constructor in the class.

Copy Constructor :

- A constructor used to initialize the state of an object similar to another object through a constructor its termed as copy constructor.
- Used in object cloning.

Object Cloning :

- Creating an identical copies of an object is referred as object cloning.

```

2
3 public class Book {
4     String title;
5
6     Book(){ // parameterless constructor or zero argument
7         this("Head First Java!");
8     }
9
10    Book(String title){ // parameterized constructor
11        this.title=title;
12    }
13
14    Book(Book b){ // copy constructor - object cloning
15        Book(b.title);
16    }
17
18 }
19

```

Constructor Overloading :

- Constructors with the same name but different parameter lists are called overloaded constructors.
- Allows creating objects in different ways.

```

class Student {
    String name;
    int age;
    String course;

    // Constructor 1 - No arguments
    Student() {
        name = "Unknown";
        age = 0;
        course = "Not Assigned";
    }

    // Constructor 2 - One argument
    Student(String n) {
        name = n;
        age = 0;
        course = "Not Assigned";
    }
}

```

```

}

// Constructor 3 - Two arguments
Student(String n, int a) {
    name = n;
    age = a;
    course = "Not Assigned";
}
}

```

Constructor Chaining :

- Calling one constructor from another in the same class using `this()`.

Function Chaining :

- A chain of successive function calls is termed as Function Chaining.

Significance of 'this' reference :

- Resolves naming conflicts between instance variables and local variables.
- Can be used to call another constructor (`this()`).

Notes :

- **A zero argument (or) parameterless constructor is applied implicitly only in the absence of any constructor in the class.**
- **A parameterless constructor is applied by default only in the absence of any constructor in a class.**
- **It's perfectly invoke a normal method from a constructor.**
(bad programming practice)
- **A constructor can't be invoked like a normal method when required.**

- **‘ this() ‘ call if present must appear as a first state in a constructor.**
- **It’s perfectly legal for a normal method to share the class name.
(bad programming practice)**